

# 1000 AI Projects — Foundation to GenAI & Agentic AI

LLM · UI (React / Streamlit) · Backend (Python FastAPI) · AI Frameworks (LangGraph / LangChain / AutoGen / CrewAI / LlamaIndex / DSPy) · Vector DB & Infra

**Master in AI Engineering** — Build production-grade GenAI, Agentic AI, RAG & MLOps systems.  
From LLM foundations to multi-agent orchestration — the complete hands-on path for AI engineers.

**Created By** Balawant M Kadam  
**Exported:** 20/9/2026, 9:05:04 pm  
**Projects in export:** 150 of 1000  
**Filters:** Level: beginner | Type: All | Domain: All | Framework: All

#0121

TRADITIONAL

BEGINNER

## Language Learning Q&A Bot

Education

Create a working Q&A Bot with LangChain for language learning using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
| LLM            | Claude 3.5 Haiku                                                            |
| UI             | Streamlit + Session State                                                   |
| BACKEND        | Python · FastAPI + Docker                                                   |
| AI FRAMEWORKS  | <a href="#">LangChain</a> <a href="#">ChromaDB</a> <a href="#">Pydantic</a> |
| VECTOR / INFRA | ChromaDB · Elasticsearch                                                    |

#0122

TRADITIONAL

BEGINNER

## Predictive Maintenance Summarizer

Manufacturing

Create a working Summarizer with LlamaIndex for predictive maintenance using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| LLM            | Gemini 1.5 Flash                                                               |
| UI             | React + Vite (basic)                                                           |
| BACKEND        | Python · FastAPI + Uvicorn                                                     |
| AI FRAMEWORKS  | <a href="#">LlamaIndex</a> <a href="#">FAISS</a> <a href="#">OpenTelemetry</a> |
| VECTOR / INFRA | Pinecone · DuckDB                                                              |

#0123

TRADITIONAL

BEGINNER

## Quality Inspection Recommender

Manufacturing

Create a working Recommender with Hugging Face Transformers for quality inspection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Weaviate · LanceDB

#0124

TRADITIONAL

BEGINNER

## Supply Chain Planning Analyzer

Manufacturing

Create a working Analyzer with LangChain for supply chain planning using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Qdrant · OpenSearch

#0125

TRADITIONAL

BEGINNER

## Robotic Process Control Assistant

Manufacturing

Create a working Assistant with LlamaIndex for robotic process control using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Milvus · FAISS

#0126

TRADITIONAL

BEGINNER

## Defect Detection Search Engine

Manufacturing

Create a working Search Engine with Hugging Face Transformers for defect detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA pgvector · ChromaDB

#0127

TRADITIONAL

BEGINNER

## Factory Digital Twin Alert System

Manufacturing

Create a working Alert System with LangChain for factory digital twin using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Manufacturing demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Neo4j · Pinecone

#0128

TRADITIONAL

BEGINNER

## Route Optimization Report Generator

Logistics &amp; Supply Chain

Create a working Report Generator with LlamaIndex for route optimization using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0129

TRADITIONAL

BEGINNER

## Fleet Management Q&A Bot

Logistics &amp; Supply Chain

Create a working Q&A Bot with Hugging Face Transformers for fleet management using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Polars](#) [Gradio](#)

VECTOR / INFRA DuckDB · Qdrant

#0130

TRADITIONAL

BEGINNER

## Warehouse Robotics Summarizer

Logistics &amp; Supply Chain

Create a working Summarizer with LangChain for warehouse robotics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA LanceDB · Milvus

#0131

TRADITIONAL

BEGINNER

## Last-Mile Delivery Recommender

Logistics &amp; Supply Chain

Create a working Recommender with LlamaIndex for last-mile delivery using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA OpenSearch · pgvector

#0132

TRADITIONAL

BEGINNER

## Freight Document Processing Analyzer

Logistics &amp; Supply Chain

Create a working Analyzer with Hugging Face Transformers for freight document processing using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA FAISS · Neo4j

#0133

TRADITIONAL

BEGINNER

## Inventory Replenishment Assistant

Logistics &amp; Supply Chain

Create a working Assistant with LangChain for inventory replenishment using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Logistics & Supply Chain demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0134

TRADITIONAL

BEGINNER

## Threat Detection Search Engine

Cybersecurity

Create a working Search Engine with LlamaIndex for threat detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Pinecone · DuckDB

#0135

TRADITIONAL

BEGINNER

## Phishing Analysis Alert System

Cybersecurity

Create a working Alert System with Hugging Face Transformers for phishing analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0136

TRADITIONAL

BEGINNER

## Vulnerability Triage Report Generator

Cybersecurity

Create a working Report Generator with LangChain for vulnerability triage using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Qdrant · OpenSearch

#0137

TRADITIONAL

BEGINNER

## Security Log Analysis Q&A Bot

Cybersecurity

Create a working Q&A Bot with LlamaIndex for security log analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Milvus · FAISS

#0138

TRADITIONAL

BEGINNER

## Incident Response Summarizer

Cybersecurity

Create a working Summarizer with Hugging Face Transformers for incident response using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA pgvector · ChromaDB

#0139

TRADITIONAL

BEGINNER

## Malware Classification Recommender

Cybersecurity

Create a working Recommender with LangChain for malware classification using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Cybersecurity demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Neo4j · Pinecone

#0140

TRADITIONAL

BEGINNER

## Campaign Copy Generation Analyzer

Marketing &amp; Advertising

Create a working Analyzer with LlamaIndex for campaign copy generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0141

TRADITIONAL

BEGINNER

## Audience Segmentation Assistant

Marketing & Advertising

Create a working Assistant with Hugging Face Transformers for audience segmentation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA DuckDB · Qdrant

#0142

TRADITIONAL

BEGINNER

## Ad Creative Optimization Search Engine

Marketing & Advertising

Create a working Search Engine with LangChain for ad creative optimization using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA LanceDB · Milvus

#0143

TRADITIONAL

BEGINNER

## SEO Content Planning Alert System

Marketing & Advertising

Create a working Alert System with LlamaIndex for seo content planning using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA OpenSearch · pgvector

#0144

TRADITIONAL

BEGINNER

## Social Listening Report Generator

Marketing &amp; Advertising

Create a working Report Generator with Hugging Face Transformers for social listening using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Gradio](#) [Polars](#)

VECTOR / INFRA FAISS · Neo4j

#0145

TRADITIONAL

BEGINNER

## Attribution Modeling Q&A Bot

Marketing &amp; Advertising

Create a working Q&A Bot with LangChain for attribution modeling using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Marketing & Advertising demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0146

TRADITIONAL

BEGINNER

## Ticket Classification Summarizer

Customer Support

Create a working Summarizer with LlamaIndex for ticket classification using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Pinecone · DuckDB

#0147

TRADITIONAL

BEGINNER

## Virtual Support Agent Recommender

Customer Support

Create a working Recommender with Hugging Face Transformers for virtual support agent using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Weaviate · LanceDB

#0148

TRADITIONAL

BEGINNER

## Sentiment Routing Analyzer

Customer Support

Create a working Analyzer with LangChain for sentiment routing using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Qdrant · OpenSearch

#0149

TRADITIONAL

BEGINNER

## Knowledge Base Search Assistant

Customer Support

Create a working Assistant with LlamaIndex for knowledge base search using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Milvus · FAISS

#0150

TRADITIONAL

BEGINNER

## Escalation Prediction Search Engine

Customer Support

Create a working Search Engine with Hugging Face Transformers for escalation prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA pgvector · ChromaDB

#0151

TRADITIONAL

BEGINNER

## Voice Support Bot Alert System

Customer Support

Create a working Alert System with LangChain for voice support bot using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Customer Support demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Neo4j · Pinecone

#0152

TRADITIONAL

BEGINNER

## Contract Review Report Generator

Legal & Compliance

Create a working Report Generator with LlamaIndex for contract review using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0153

TRADITIONAL

BEGINNER

## Legal Research Q&A Bot

Legal &amp; Compliance

Create a working Q&A Bot with Hugging Face Transformers for legal research using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA DuckDB · Qdrant

#0154

TRADITIONAL

BEGINNER

## Policy Compliance Summarizer

Legal &amp; Compliance

Create a working Summarizer with LangChain for policy compliance using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Gradio](#) [Polars](#)

VECTOR / INFRA LanceDB · Milvus

#0155

TRADITIONAL

BEGINNER

## E-Discovery Recommender

Legal &amp; Compliance

Create a working Recommender with LlamaIndex for e-discovery using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA OpenSearch · pgvector

#0156

TRADITIONAL

BEGINNER

## Clause Extraction Analyzer

Legal &amp; Compliance

Create a working Analyzer with Hugging Face Transformers for clause extraction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA FAISS · Neo4j

#0157

TRADITIONAL

BEGINNER

## Regulatory Monitoring Assistant

Legal &amp; Compliance

Create a working Assistant with LangChain for regulatory monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Legal & Compliance demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0158

TRADITIONAL

BEGINNER

## Resume Screening Search Engine

Human Resources

Create a working Search Engine with LlamaIndex for resume screening using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Pinecone · DuckDB

#0159

TRADITIONAL

BEGINNER

## Interview Assistant Alert System

Human Resources

Create a working Alert System with Hugging Face Transformers for interview assistant using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Weaviate · LanceDB

#0160

TRADITIONAL

BEGINNER

## Employee Onboarding Report Generator

Human Resources

Create a working Report Generator with LangChain for employee onboarding using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Qdrant · OpenSearch

#0161

TRADITIONAL

BEGINNER

## Attrition Prediction Q&A Bot

Human Resources

Create a working Q&A Bot with LlamaIndex for attrition prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Milvus · FAISS

#0162

TRADITIONAL

BEGINNER

## Skills Gap Analysis Summarizer

Human Resources

Create a working Summarizer with Hugging Face Transformers for skills gap analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA pgvector · ChromaDB

#0163

TRADITIONAL

BEGINNER

## Performance Review Summarizer Recommender

Human Resources

Create a working Recommender with LangChain for performance review summarizer using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Human Resources demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Neo4j · Pinecone

#0164

TRADITIONAL

BEGINNER

## Crop Disease Detection Analyzer

Agriculture

Create a working Analyzer with LlamaIndex for crop disease detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0165

TRADITIONAL

BEGINNER

## Yield Prediction Assistant

Agriculture

Create a working Assistant with Hugging Face Transformers for yield prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA DuckDB · Qdrant

#0166

TRADITIONAL

BEGINNER

## Precision Irrigation Search Engine

Agriculture

Create a working Search Engine with LangChain for precision irrigation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA LanceDB · Milvus

#0167

TRADITIONAL

BEGINNER

## Pest Monitoring Alert System

Agriculture

Create a working Alert System with LlamaIndex for pest monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA OpenSearch · pgvector

#0168

TRADITIONAL

BEGINNER

## Soil Health Analytics Report Generator

Agriculture

Create a working Report Generator with Hugging Face Transformers for soil health analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA FAISS · Neo4j

#0169

TRADITIONAL

BEGINNER

## Farm Drone Imagery Q&A Bot

Agriculture

Create a working Q&A Bot with LangChain for farm drone imagery using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Agriculture demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Polars](#) [Gradio](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0170

TRADITIONAL

BEGINNER

## Grid Load Forecasting Summarizer

Energy &amp; Utilities

Create a working Summarizer with LlamaIndex for grid load forecasting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Pinecone · DuckDB

#0171

TRADITIONAL

BEGINNER

## Renewable Output Prediction Recommender

Energy &amp; Utilities

Create a working Recommender with Hugging Face Transformers for renewable output prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Weaviate · LanceDB

#0172

TRADITIONAL

BEGINNER

## Asset Health Monitoring Analyzer

Energy &amp; Utilities

Create a working Analyzer with LangChain for asset health monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Qdrant · OpenSearch

#0173

TRADITIONAL

BEGINNER

## Energy Trading Assistant

Energy &amp; Utilities

Create a working Assistant with LlamaIndex for energy trading using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Milvus · FAISS

#0174

TRADITIONAL

BEGINNER

## Outage Detection Search Engine

Energy &amp; Utilities

Create a working Search Engine with Hugging Face Transformers for outage detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Gradio](#) [Polars](#)

VECTOR / INFRA pgvector · ChromaDB

#0175

TRADITIONAL

BEGINNER

## Smart Meter Analytics Alert System

Energy &amp; Utilities

Create a working Alert System with LangChain for smart meter analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Energy & Utilities demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Neo4j · Pinecone

#0176

TRADITIONAL

BEGINNER

## Citizen Service Assistant Report Generator

Government &amp; Public Sector

Create a working Report Generator with LlamaIndex for citizen service assistant using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0177

TRADITIONAL

BEGINNER

## Document Processing Q&A Bot

Government &amp; Public Sector

Create a working Q&A Bot with Hugging Face Transformers for document processing using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA DuckDB · Qdrant

#0178

TRADITIONAL

BEGINNER

## Policy Impact Analysis Summarizer

Government &amp; Public Sector

Create a working Summarizer with LangChain for policy impact analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA LanceDB · Milvus

#0179

TRADITIONAL

BEGINNER

## Emergency Response Recommender

Government &amp; Public Sector

Create a working Recommender with LlamaIndex for emergency response using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Polars](#) [Gradio](#)

VECTOR / INFRA OpenSearch · pgvector

#0180

TRADITIONAL

BEGINNER

## Permit Review Analyzer

Government &amp; Public Sector

Create a working Analyzer with Hugging Face Transformers for permit review using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA FAISS · Neo4j

#0181

TRADITIONAL

BEGINNER

## Public Feedback Mining Assistant

Government &amp; Public Sector

Create a working Assistant with LangChain for public feedback mining using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Government & Public Sector demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0182

TRADITIONAL

BEGINNER

## Content Recommendation Search Engine

Media &amp; Entertainment

Create a working Search Engine with LlamaIndex for content recommendation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Pinecone · DuckDB

#0183

TRADITIONAL

BEGINNER

## Script Generation Alert System

Media &amp; Entertainment

Create a working Alert System with Hugging Face Transformers for script generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Weaviate · LanceDB

#0184

TRADITIONAL

BEGINNER

## Video Editing Assistant Report Generator

Media &amp; Entertainment

Create a working Report Generator with LangChain for video editing assistant using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Qdrant · OpenSearch

#0185

TRADITIONAL

BEGINNER

## Music Generation Q&A Bot

Media &amp; Entertainment

Create a working Q&A Bot with LlamaIndex for music generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Milvus · FAISS

#0186

TRADITIONAL

BEGINNER

## Subtitle Localization Summarizer

Media &amp; Entertainment

Create a working Summarizer with Hugging Face Transformers for subtitle localization using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA pgvector · ChromaDB

#0187

TRADITIONAL

BEGINNER

## Content Moderation Recommender

Media &amp; Entertainment

Create a working Recommender with LangChain for content moderation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Media & Entertainment demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Neo4j · Pinecone

#0188

TRADITIONAL

BEGINNER

## NPC Dialogue Systems Analyzer

Gaming

Create a working Analyzer with LlamaIndex for npc dialogue systems using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0189

TRADITIONAL

BEGINNER

## Procedural Level Generation Assistant

Gaming

Create a working Assistant with Hugging Face Transformers for procedural level generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Polars](#) [Gradio](#)

VECTOR / INFRA DuckDB · Qdrant

#0190

TRADITIONAL

BEGINNER

## Player Behavior Analytics Search Engine

Gaming

Create a working Search Engine with LangChain for player behavior analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA LanceDB · Milvus

#0191

TRADITIONAL

BEGINNER

## Matchmaking Alert System

Gaming

Create a working Alert System with LlamaIndex for matchmaking using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA OpenSearch · pgvector

#0192

TRADITIONAL

BEGINNER

## Anti-Cheat Detection Report Generator

Gaming

Create a working Report Generator with Hugging Face Transformers for anti-cheat detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA FAISS · Neo4j

#0193

TRADITIONAL

BEGINNER

## Game Testing Bots Q&A Bot

Gaming

Create a working Q&A Bot with LangChain for game testing bots using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Gaming demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0194

TRADITIONAL

BEGINNER

## Pick-and-Place Control Summarizer

Robotics

Create a working Summarizer with LlamaIndex for pick-and-place control using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Pinecone · DuckDB

#0195

TRADITIONAL

BEGINNER

## SLAM Navigation Recommender

Robotics

Create a working Recommender with Hugging Face Transformers for slam navigation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0196

TRADITIONAL

BEGINNER

## Manipulation Planning Analyzer

Robotics

Create a working Analyzer with LangChain for manipulation planning using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Qdrant · OpenSearch

#0197

TRADITIONAL

BEGINNER

## Human-Robot Collaboration Assistant

Robotics

Create a working Assistant with LlamaIndex for human-robot collaboration using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Milvus · FAISS

#0198

TRADITIONAL

BEGINNER

## Swarm Coordination Search Engine

Robotics

Create a working Search Engine with Hugging Face Transformers for swarm coordination using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

|                |                                                                                            |
|----------------|--------------------------------------------------------------------------------------------|
| LLM            | GPT-4o mini                                                                                |
| UI             | Streamlit + FastAPI                                                                        |
| BACKEND        | Python · FastAPI + Pydantic v2                                                             |
| AI FRAMEWORKS  | <a href="#">Hugging Face Transformers</a> <a href="#">MLflow</a> <a href="#">Streamlit</a> |
| VECTOR / INFRA | pgvector · ChromaDB                                                                        |

#0199

TRADITIONAL

BEGINNER

## Sim-to-Real Transfer Alert System

Robotics

Create a working Alert System with LangChain for sim-to-real transfer using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Robotics demo with a clean UI.

|                |                                                                         |
|----------------|-------------------------------------------------------------------------|
| LLM            | Claude 3.5 Haiku                                                        |
| UI             | Streamlit + Session State                                               |
| BACKEND        | Python · FastAPI + Docker                                               |
| AI FRAMEWORKS  | <a href="#">LangChain</a> <a href="#">Polars</a> <a href="#">Gradio</a> |
| VECTOR / INFRA | Neo4j · Pinecone                                                        |

#0200

TRADITIONAL

BEGINNER

## Perception Fusion Report Generator

Autonomous Vehicles

Create a working Report Generator with LlamaIndex for perception fusion using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| LLM            | Gemini 1.5 Flash                                                          |
| UI             | React + Vite (basic)                                                      |
| BACKEND        | Python · FastAPI + Uvicorn                                                |
| AI FRAMEWORKS  | <a href="#">LlamaIndex</a> <a href="#">DuckDB</a> <a href="#">FastAPI</a> |
| VECTOR / INFRA | Elasticsearch · Weaviate                                                  |

#0201

TRADITIONAL

BEGINNER

## Trajectory Prediction Q&A Bot

Autonomous Vehicles

Create a working Q&A Bot with Hugging Face Transformers for trajectory prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA DuckDB · Qdrant

#0202

TRADITIONAL

BEGINNER

## Driver Monitoring Summarizer

Autonomous Vehicles

Create a working Summarizer with LangChain for driver monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA LanceDB · Milvus

#0203

TRADITIONAL

BEGINNER

## HD Map Generation Recommender

Autonomous Vehicles

Create a working Recommender with LlamaIndex for hd map generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA OpenSearch · pgvector

#0204

TRADITIONAL

BEGINNER

## Scenario Simulation Analyzer

Autonomous Vehicles

Create a working Analyzer with Hugging Face Transformers for scenario simulation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Gradio](#) [Polars](#)

VECTOR / INFRA FAISS · Neo4j

#0205

TRADITIONAL

BEGINNER

## Fleet Teleoperation Assistant

Autonomous Vehicles

Create a working Assistant with LangChain for fleet teleoperation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Autonomous Vehicles demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0206

TRADITIONAL

BEGINNER

## Literature Synthesis Search Engine

Scientific Research

Create a working Search Engine with LlamaIndex for literature synthesis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Pinecone · DuckDB

#0207

TRADITIONAL

BEGINNER

## Protein Structure Prediction Alert System

Scientific Research

Create a working Alert System with Hugging Face Transformers for protein structure prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Weaviate · LanceDB

#0208

TRADITIONAL

BEGINNER

## Experiment Design Report Generator

Scientific Research

Create a working Report Generator with LangChain for experiment design using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Qdrant · OpenSearch

#0209

TRADITIONAL

BEGINNER

## Lab Automation Q&A Bot

Scientific Research

Create a working Q&A Bot with LlamaIndex for lab automation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Milvus · FAISS

#0210

TRADITIONAL

BEGINNER

## Materials Discovery Summarizer

Scientific Research

Create a working Summarizer with Hugging Face Transformers for materials discovery using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA pgvector · ChromaDB

#0211

TRADITIONAL

BEGINNER

## Research Data Analysis Recommender

Scientific Research

Create a working Recommender with LangChain for research data analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Scientific Research demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Neo4j · Pinecone

#0212

TRADITIONAL

BEGINNER

## Emissions Monitoring Analyzer

Climate &amp; Environment

Create a working Analyzer with LlamaIndex for emissions monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0213

TRADITIONAL

BEGINNER

## Wildfire Prediction Assistant

Climate &amp; Environment

Create a working Assistant with Hugging Face Transformers for wildfire prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA DuckDB · Qdrant

#0214

TRADITIONAL

BEGINNER

## Biodiversity Tracking Search Engine

Climate &amp; Environment

Create a working Search Engine with LangChain for biodiversity tracking using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Gradio](#) [Polars](#)

VECTOR / INFRA LanceDB · Milvus

#0215

TRADITIONAL

BEGINNER

## Carbon Accounting Alert System

Climate &amp; Environment

Create a working Alert System with LlamaIndex for carbon accounting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA OpenSearch · pgvector

#0216

TRADITIONAL

BEGINNER

## Weather Forecasting Report Generator

Climate &amp; Environment

Create a working Report Generator with Hugging Face Transformers for weather forecasting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| LLM            | GPT-4o mini                                                                                 |
| UI             | Streamlit + FastAPI                                                                         |
| BACKEND        | Python · FastAPI + Pydantic v2                                                              |
| AI FRAMEWORKS  | <a href="#">Hugging Face Transformers</a> <a href="#">Pydantic</a> <a href="#">ChromaDB</a> |
| VECTOR / INFRA | FAISS · Neo4j                                                                               |

#0217

TRADITIONAL

BEGINNER

## Ocean Data Analytics Q&A Bot

Climate &amp; Environment

Create a working Q&A Bot with LangChain for ocean data analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Climate & Environment demo with a clean UI.

|                |                                                                               |
|----------------|-------------------------------------------------------------------------------|
| LLM            | Claude 3.5 Haiku                                                              |
| UI             | Streamlit + Session State                                                     |
| BACKEND        | Python · FastAPI + Docker                                                     |
| AI FRAMEWORKS  | <a href="#">LangChain</a> <a href="#">OpenTelemetry</a> <a href="#">FAISS</a> |
| VECTOR / INFRA | ChromaDB · Elasticsearch                                                      |

#0218

TRADITIONAL

BEGINNER

## Network Optimization Summarizer

Telecommunications

Create a working Summarizer with LlamaIndex for network optimization using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
| LLM            | Gemini 1.5 Flash                                                            |
| UI             | React + Vite (basic)                                                        |
| BACKEND        | Python · FastAPI + Uvicorn                                                  |
| AI FRAMEWORKS  | <a href="#">LlamaIndex</a> <a href="#">MLflow</a> <a href="#">Streamlit</a> |
| VECTOR / INFRA | Pinecone · DuckDB                                                           |

#0219

TRADITIONAL

BEGINNER

## Churn Prediction Recommender

Telecommunications

Create a working Recommender with Hugging Face Transformers for churn prediction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Weaviate · LanceDB

#0220

TRADITIONAL

BEGINNER

## Fault Diagnosis Analyzer

Telecommunications

Create a working Analyzer with LangChain for fault diagnosis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Qdrant · OpenSearch

#0221

TRADITIONAL

BEGINNER

## Customer Care Assistant Assistant

Telecommunications

Create a working Assistant with LlamaIndex for customer care assistant using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Milvus · FAISS

#0222

TRADITIONAL

BEGINNER

## 5G Slice Management Search Engine

Telecommunications

Create a working Search Engine with Hugging Face Transformers for 5g slice management using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA pgvector · ChromaDB

#0223

TRADITIONAL

BEGINNER

## Spectrum Analytics Alert System

Telecommunications

Create a working Alert System with LangChain for spectrum analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Telecommunications demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Neo4j · Pinecone

#0224

TRADITIONAL

BEGINNER

## Property Valuation Report Generator

Real Estate &amp; Construction

Create a working Report Generator with LlamaIndex for property valuation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0225

TRADITIONAL

BEGINNER

## Site Safety Monitoring Q&A Bot

Real Estate &amp; Construction

Create a working Q&A Bot with Hugging Face Transformers for site safety monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA DuckDB · Qdrant

#0226

TRADITIONAL

BEGINNER

## Lease Abstraction Summarizer

Real Estate &amp; Construction

Create a working Summarizer with LangChain for lease abstraction using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA LanceDB · Milvus

#0227

TRADITIONAL

BEGINNER

## BIM Analysis Recommender

Real Estate &amp; Construction

Create a working Recommender with LlamaIndex for bim analysis using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA OpenSearch · pgvector

#0228

TRADITIONAL

BEGINNER

## Construction Scheduling Analyzer

Real Estate &amp; Construction

Create a working Analyzer with Hugging Face Transformers for construction scheduling using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA FAISS · Neo4j

#0229

TRADITIONAL

BEGINNER

## Tenant Support Assistant

Real Estate &amp; Construction

Create a working Assistant with LangChain for tenant support using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Real Estate & Construction demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Polars](#) [Gradio](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0230

TRADITIONAL

BEGINNER

## Itinerary Planning Search Engine

Travel &amp; Hospitality

Create a working Search Engine with LlamaIndex for itinerary planning using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Pinecone · DuckDB

#0231

TRADITIONAL

BEGINNER

## Dynamic Pricing Alert System

Travel &amp; Hospitality

Create a working Alert System with Hugging Face Transformers for dynamic pricing using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA Weaviate · LanceDB

#0232

TRADITIONAL

BEGINNER

## Review Analytics Report Generator

Travel &amp; Hospitality

Create a working Report Generator with LangChain for review analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Qdrant · OpenSearch

#0233

TRADITIONAL

BEGINNER

## Virtual Concierge Q&A Bot

Travel &amp; Hospitality

Create a working Q&A Bot with LlamaIndex for virtual concierge using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Milvus · FAISS

#0234

TRADITIONAL

BEGINNER

## Demand Forecasting Summarizer

Travel &amp; Hospitality

Create a working Summarizer with Hugging Face Transformers for demand forecasting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Gradio](#) [Polars](#)

VECTOR / INFRA pgvector · ChromaDB

#0235

TRADITIONAL

BEGINNER

## Multilingual Support Recommender

Travel &amp; Hospitality

Create a working Recommender with LangChain for multilingual support using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Travel & Hospitality demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Neo4j · Pinecone

#0236

TRADITIONAL

BEGINNER

## Claims Automation Analyzer

Insurance

Create a working Analyzer with LlamaIndex for claims automation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0237

TRADITIONAL

BEGINNER

## Risk Assessment Assistant

Insurance

Create a working Assistant with Hugging Face Transformers for risk assessment using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA DuckDB · Qdrant

#0238

TRADITIONAL

BEGINNER

## Policy Comparison Search Engine

Insurance

Create a working Search Engine with LangChain for policy comparison using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA LanceDB · Milvus

#0239

TRADITIONAL

BEGINNER

## Damage Estimation Alert System

Insurance

Create a working Alert System with LlamaIndex for damage estimation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Polars](#) [Gradio](#)

VECTOR / INFRA OpenSearch · pgvector

#0240

TRADITIONAL

BEGINNER

## Fraud Detection Report Generator

Insurance

Create a working Report Generator with Hugging Face Transformers for fraud detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA FAISS · Neo4j

#0241

TRADITIONAL

BEGINNER

## Underwriting Copilot Q&A Bot

Insurance

Create a working Q&A Bot with LangChain for underwriting copilot using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Insurance demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0242

TRADITIONAL

BEGINNER

## Code Review Summarizer

Software Engineering

Create a working Summarizer with LlamaIndex for code review using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA Pinecone · DuckDB

#0243

TRADITIONAL

BEGINNER

## Test Generation Recommender

Software Engineering

Create a working Recommender with Hugging Face Transformers for test generation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA Weaviate · LanceDB

#0244

TRADITIONAL

BEGINNER

## Incident Root Cause Analyzer

Software Engineering

Create a working Analyzer with LangChain for incident root cause using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Qdrant · OpenSearch

#0245

TRADITIONAL

BEGINNER

## Documentation Assistant

Software Engineering

Create a working Assistant with LlamaIndex for documentation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Milvus · FAISS

#0246

TRADITIONAL

BEGINNER

## Legacy Migration Search Engine

Software Engineering

Create a working Search Engine with Hugging Face Transformers for legacy migration using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| LLM            | GPT-4o mini                                                                                 |
| UI             | Streamlit + FastAPI                                                                         |
| BACKEND        | Python · FastAPI + Pydantic v2                                                              |
| AI FRAMEWORKS  | <a href="#">Hugging Face Transformers</a> <a href="#">Pydantic</a> <a href="#">ChromaDB</a> |
| VECTOR / INFRA | pgvector · ChromaDB                                                                         |

#0247

TRADITIONAL

BEGINNER

## Developer Copilot Alert System

Software Engineering

Create a working Alert System with LangChain for developer copilot using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Software Engineering demo with a clean UI.

|                |                                                                               |
|----------------|-------------------------------------------------------------------------------|
| LLM            | Claude 3.5 Haiku                                                              |
| UI             | Streamlit + Session State                                                     |
| BACKEND        | Python · FastAPI + Docker                                                     |
| AI FRAMEWORKS  | <a href="#">LangChain</a> <a href="#">OpenTelemetry</a> <a href="#">FAISS</a> |
| VECTOR / INFRA | Neo4j · Pinecone                                                              |

#0248

TRADITIONAL

BEGINNER

## Clinical Notes Report Generator

Healthcare

Create a working Report Generator with LlamaIndex for clinical notes using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
| LLM            | Gemini 1.5 Flash                                                            |
| UI             | React + Vite (basic)                                                        |
| BACKEND        | Python · FastAPI + Uvicorn                                                  |
| AI FRAMEWORKS  | <a href="#">LlamaIndex</a> <a href="#">MLflow</a> <a href="#">Streamlit</a> |
| VECTOR / INFRA | Elasticsearch · Weaviate                                                    |

#0249

TRADITIONAL

BEGINNER

## Diagnostic Imaging Q&A Bot

Healthcare

Create a working Q&A Bot with Hugging Face Transformers for diagnostic imaging using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [Polars](#) [Gradio](#)

VECTOR / INFRA DuckDB · Qdrant

#0250

TRADITIONAL

BEGINNER

## Patient Triage Summarizer

Healthcare

Create a working Summarizer with LangChain for patient triage using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA LanceDB · Milvus

#0251

TRADITIONAL

BEGINNER

## Drug Interaction Checker Recommender

Healthcare

Create a working Recommender with LlamaIndex for drug interaction checker using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA OpenSearch · pgvector

#0252

TRADITIONAL

BEGINNER

## EHR Records Analyzer

Healthcare

Create a working Analyzer with Hugging Face Transformers for ehr records using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA FAISS · Neo4j

#0253

TRADITIONAL

BEGINNER

## Remote Patient Monitoring Assistant

Healthcare

Create a working Assistant with LangChain for remote patient monitoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Healthcare demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0254

TRADITIONAL

BEGINNER

## Fraud Detection Search Engine

Finance &amp; Banking

Create a working Search Engine with LlamaIndex for fraud detection using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Gradio](#) [Polars](#)

VECTOR / INFRA Pinecone · DuckDB

#0255

TRADITIONAL

BEGINNER

## Credit Scoring Alert System

Finance &amp; Banking

Create a working Alert System with Hugging Face Transformers for credit scoring using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0256

TRADITIONAL

BEGINNER

## Portfolio Analytics Report Generator

Finance &amp; Banking

Create a working Report Generator with LangChain for portfolio analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Qdrant · OpenSearch

#0257

TRADITIONAL

BEGINNER

## Loan Underwriting Q&A Bot

Finance &amp; Banking

Create a working Q&A Bot with LlamaIndex for loan underwriting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Milvus · FAISS

#0258

TRADITIONAL

BEGINNER

## Transaction Reconciliation Summarizer

Finance &amp; Banking

Create a working Summarizer with Hugging Face Transformers for transaction reconciliation using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA pgvector · ChromaDB

#0259

TRADITIONAL

BEGINNER

## Market Sentiment Recommender

Finance &amp; Banking

Create a working Recommender with LangChain for market sentiment using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Finance & Banking demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Neo4j · Pinecone

#0260

TRADITIONAL

BEGINNER

## Product Recommendations Analyzer

Retail &amp; E-commerce

Create a working Analyzer with LlamaIndex for product recommendations using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [DuckDB](#) [FastAPI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0261

TRADITIONAL

BEGINNER

## Demand Forecasting Assistant

Retail &amp; E-commerce

Create a working Assistant with Hugging Face Transformers for demand forecasting using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [ChromaDB](#) [Pydantic](#)

VECTOR / INFRA DuckDB · Qdrant

#0262

TRADITIONAL

BEGINNER

## Visual Search Search Engine

Retail &amp; E-commerce

Create a working Search Engine with LangChain for visual search using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [FAISS](#) [OpenTelemetry](#)

VECTOR / INFRA LanceDB · Milvus

#0263

TRADITIONAL

BEGINNER

## Dynamic Pricing Alert System (v4)

Retail &amp; E-commerce

Create a working Alert System with LlamaIndex for dynamic pricing using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Streamlit](#) [MLflow](#)

VECTOR / INFRA OpenSearch · pgvector

#0264

TRADITIONAL

BEGINNER

## Cart Abandonment Report Generator

Retail &amp; E-commerce

Create a working Report Generator with Hugging Face Transformers for cart abandonment using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM GPT-4o mini

UI Streamlit + FastAPI

BACKEND Python · FastAPI + Pydantic v2

AI FRAMEWORKS [Hugging Face Transformers](#) [Gradio](#) [Polars](#)

VECTOR / INFRA FAISS · Neo4j

#0265

TRADITIONAL

BEGINNER

## Inventory Optimization Q&A Bot

Retail &amp; E-commerce

Create a working Q&A Bot with LangChain for inventory optimization using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Retail & E-commerce demo with a clean UI.

LLM Claude 3.5 Haiku

UI Streamlit + Session State

BACKEND Python · FastAPI + Docker

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [DuckDB](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0266

TRADITIONAL

BEGINNER

## Adaptive Learning Paths Summarizer

Education

Create a working Summarizer with LlamaIndex for adaptive learning paths using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

LLM Gemini 1.5 Flash

UI React + Vite (basic)

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Pinecone · DuckDB

#0267

TRADITIONAL

BEGINNER

## Essay Grading Recommender

Education

Create a working Recommender with Hugging Face Transformers for essay grading using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

LLM Llama 3.1 8B

UI Streamlit Chat UI

BACKEND Python · FastAPI + SQLite + SQLAlchemy

AI FRAMEWORKS [Hugging Face Transformers](#) [OpenTelemetry](#) [FAISS](#)

VECTOR / INFRA Weaviate · LanceDB

#0268

TRADITIONAL

BEGINNER

## Tutoring Assistant Analyzer

Education

Create a working Analyzer with LangChain for tutoring assistant using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

LLM Qwen 2.5 7B

UI Streamlit

BACKEND Python · FastAPI + BackgroundTasks

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Streamlit](#)

VECTOR / INFRA Qdrant · OpenSearch

#0269

TRADITIONAL

BEGINNER

## Curriculum Generator Assistant

Education

Create a working Assistant with LlamaIndex for curriculum generator using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

LLM Mistral Small

UI Gradio

BACKEND Python · FastAPI + Redis cache

AI FRAMEWORKS [LlamaIndex](#) [Polars](#) [Gradio](#)

VECTOR / INFRA Milvus · FAISS

# Student Engagement Analytics Search Engine

Education

Create a working Search Engine with Hugging Face Transformers for student engagement analytics using prompt engineering, structured outputs and basic retrieval, wrapped in a simple Education demo with a clean UI.

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| LLM            | GPT-4o mini                                                               |
| UI             | Streamlit + FastAPI                                                       |
| BACKEND        | Python · FastAPI + Pydantic v2                                            |
| AI FRAMEWORKS  | <div>Hugging Face Transformers</div> <div>DuckDB</div> <div>FastAPI</div> |
| VECTOR / INFRA | pgvector · ChromaDB                                                       |

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